

Platform Configuration, Environment Setup & Channel Integrations Manual

Comprehensive Technical Specification & Step-by-Step Operator Guide covering Environment Variables (.env), Cryptographic Vault (AES-256-GCM), Multi-Tenant RBAC, and External APIs (WhatsApp, Instagram, Twilio, Meta & Google Ads).

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Document Roadmap & Table of Contents

Section	Topic & Coverage	Page
Section 1	Architecture Blueprint, Multi-Tenancy & Operational Scope	Page 2
Section 2	Infrastructure & Server Environment Variables (.env)	Page 3
Section 3	In-App Client Integrations Hub (Centralized UI Guide)	Page 4
Section 4	Step-by-Step Channel Integrations (WhatsApp, Instagram, Twilio, Email)	Pages 5 - 6
Section 5	Advertising & Lead Ingestion Setup (Meta Ads & Google Ads)	Page 7
Section 6	Local Webhook Development, Tunnels & Webhook Verification	Page 8
Section 7	Security Protocols, AES-256 Secret Encryption & RBAC Matrix	Page 9
Section 8	Troubleshooting, Error Diagnostics & Go-Live Checklist	Page 10

1. Architecture Blueprint & Operational Scope

- FlumenX ConectOS operates as a multi-tenant digital marketing operations platform. To balance high security, zero cross-tenant credential leakage, and operational flexibility, system configuration is split into two distinct tiers:
- **Tier 1: Global Infrastructure & System Secrets (.env):** Managed exclusively by platform DevOps and agency administrators. These variables control database clustering, session JWT signing, server port bindings, and the master cryptographic symmetric key.
 - **Tier 2: Tenant-Level Channel Credentials (In-App Integrations Hub):** Configured per client workspace directly inside the UI. Each client can independently register and manage their own WhatsApp Business account, Instagram DMs, Twilio SMS numbers, and Meta/Google ad accounts without developer intervention.

End-to-End Cryptographic Dataflow

Step	System Component	Functional Action & Security Boundary
1	Client User / Admin	Submits API keys or tokens in the Integrations Hub (e.g. WhatsApp permanent token, Twilio Auth Token).
2	Next.js Frontend	Transmits credentials over TLS/HTTPS with JWT authorization bearer header to Express API gateway.
3	Express REST Gateway	Validates tenant membership via RBAC middleware; invokes EncryptionService.
4	AES-256-GCM Vault	Encrypts secret using master ENCRYPTION_SECRET_KEY with unique random IV and authentication tag.
5	MongoDB Collection	Stores encrypted ciphertext in communicationproviders tagged with clientId.
6	Outbound Dispatcher	When dispatching a lead message or syncing ads, token is decrypted in-memory and dispatched to external API.

TENANT ISOLATION GUARANTEE

Every database query for communication channels, conversations, leads, and ad metrics is automatically scoped by the user's active clientId. Cross-tenant access is strictly blocked by RBAC middleware, preventing data leaks across clients.

2. Infrastructure & Environment Variables (.env)

The backend runtime validates all environment variables against a strict Zod schema in backend/src/config/env.ts. Missing or malformed required variables prevent application boot:

Variable Name	Default / Example	Req.	Description & Operational Function
PORT	5000	Opt	TCP port on which Express listens for incoming HTTP and webhook requests.
NODE_ENV	development	Opt	Runtime environment mode: development, production, or test.
MONGODB_URI	mongodb://localhost:27017/flumenx_connect_os	YES	MongoDB connection string. In production, use MongoDB Atlas replica set URI with TLS enabled.
JWT_SECRET	Min 32-char random string	YES	Cryptographic secret key used by HMAC-SHA256 to sign session authentication tokens.
JWT_EXPIRES_IN	7d	Opt	Session token lifespan before re-authentication is enforced (e.g. 24h, 7d).
ENCRYPTION_SECRET_KEY	64-hex char string (32 bytes)	YES	Master symmetric encryption key utilized by the AES-256-GCM vault to encrypt tokens in MongoDB.
CORS_ORIGIN	http://localhost:3000	YES	Allowed origin for CORS. In production, set to live domain (e.g. https://app.yourdomain.com).
INITIAL_ADMIN_EMAIL	admin@flumenx.com	YES	Seed email address for platform provisioning on fresh MongoDB initialization.
INITIAL_ADMIN_PASSWORD	Min 8 characters	YES	Seed password for initial Super Administrator bootstrap.

Frontend Environment Variables (frontend/.env.local)

The Next.js 14 client application reads client-exposed variables prefixed with NEXT_PUBLIC_ during build and client hydration:

Variable Name	Local Dev Value	Production Recommendation
NEXT_PUBLIC_API_URL	http://localhost:5000/api/v1	Set to your public HTTPS API endpoint, e.g. https://api.flumenx.yourdomain.com/api/v1

Generating Production Cryptographic Keys

Execute either of the following CLI commands to generate a random 64-hex character encryption key for ENCRYPTION_SECRET_KEY:

```
# Option A: Using Node.js crypto module (recommended):
node -e "console.log(require('crypto').randomBytes(32).toString('hex'))"

# Option B: Using OpenSSL CLI:
openssl rand -hex 32
```

3. In-App Client Integrations Hub (Centralized UI)

To configure channels, ad accounts, and webhooks, navigate in the browser to: **Client Workspace > Integrations Hub** (/client/integrations).

All configuration forms feature interactive **(i) Information Icons** providing inline guidance, developer console URLs, required permission scopes, and step-by-step token generation instructions.

Hub Tab	Supported Providers & Platforms	Key Capabilities
1. Messaging & Channels	- WhatsApp Cloud API - Instagram Direct Messaging - Facebook Messenger - Twilio 2-Way SMS Gateway - Resend / Custom SMTP Email	Enables sending and receiving messages in the Unified Inbox, CRM Lead Messaging modal, and automated workflow auto-replies. Supports testing connection and toggling default outbound channel.
2. Advertising & Campaigns	- Meta Marketing API (Facebook/Instagram Ads) - Google Ads MCC / Customer Account	Automated sync of ad spend, impressions, clicks, CPC, ROAS, and instant lead capture from Meta Instant Lead Forms directly into CRM pipeline.
3. Webhook & Callback URLs	- Inbound Messaging Webhook - Webhook Verify Tokens - Lead Intake Ingestion Webhook	Ready-to-copy endpoints for Meta App Dashboard, Twilio Webhook settings, and website forms. Displays active verify tokens with copy-to-clipboard functionality.

Credential Security & Secret Masking

When credentials are submitted via the Integrations UI:

1. The backend encrypts access tokens and passwords using AES-256-GCM with unique initialization vectors (IV) before writing to the communicationproviders collection.
2. When credentials are sent back to the frontend, sensitive fields (tokens, secrets, passwords) are masked as to prevent exposure via browser inspector or network sniffing.
3. When testing connections via the **Test Connection** button, the backend decrypts tokens in-memory, executes a lightweight ping to the external provider API, and reports connection status (HTTP 200 / valid authentication).

LEAD MESSAGING CRM SHORTCUT

Configuring a default messaging provider (e.g. WhatsApp or Twilio) activates the direct 'Message' action button inside the Leads CRM table (/client/leads) and Lead Detail View (/client/leads/[leadId]), allowing agents to trigger WhatsApp Web, send SMS, or launch the Unified Inbox with 1 click.

4. Step-by-Step Channel Integrations Guide

4.1 WhatsApp Cloud API (Meta for Developers)

Connect official WhatsApp Business numbers to send outbound messages, manage incoming chats, and execute automated CRM lead follow-ups.

Step	Action Required	Console Location & Detailed Instructions
1	Create Meta App	Go to developers.facebook.com > My Apps > Create App . Select app type: Business . Name the app after your agency or client.
2	Add WhatsApp Product	In App Dashboard, click Add Product and select WhatsApp . Complete business onboarding and add a payment method if sending marketing templates.
3	Retrieve Phone & WABA IDs	Navigate to WhatsApp > API Setup . Copy the numeric Phone Number ID (e.g. 104928374829102) and WhatsApp Business Account ID (WABA ID).
4	Generate Permanent Token	Temporary tokens expire in 24h. For permanent access: Go to business.facebook.com > Business Settings > System Users . Click Add (Role: Admin). Click Generate New Token , select your App, and enable permissions: <code>whatsapp_business_messaging</code> and <code>whatsapp_business_management</code> .
5	Configure Webhook	In Meta App Dashboard > WhatsApp > Configuration > Edit Webhook : • Callback URL: <code>https://your-domain.com/api/v1/conversations/webhooks/whatsapp</code> • Verify Token: Paste your chosen Verify Token (e.g. <code>flumenx_wa_verify_2026</code>) • Subscribed Fields: Click Manage and check messages.
6	Save in FlumenX Hub	Paste Phone Number ID, WABA ID, and Permanent Access Token into Integrations Hub > WhatsApp . Click Test Connection , then Save Channel .

4.2 Instagram Direct Messaging & Facebook Messenger

Manage Direct Messages (DMs) from Instagram and Messenger inside the Unified Inbox alongside CRM records.

Step	Requirement	Instructions & Permissions
1	Professional Account	Ensure the Instagram account is set to Creator or Business and linked to the official Facebook Page in Meta Business Suite.
2	Message Access Permission	In Instagram mobile app: Settings > Privacy > Messages > Connected Tools : Toggle Allow Access to Messages to ON.
3	Token Permissions	Generate a Page Access Token with: <code>instagram_basic</code> , <code>instagram_manage_messages</code> , <code>pages_manage_metadata</code> , <code>pages_read_engagement</code> .
4	Webhook Subscriptions	In Meta App Dashboard > Webhooks > Select Instagram : Subscribe to messages and messaging_postbacks .

4.3 Twilio Two-Way SMS Gateway

Enables text messaging to CRM leads and automated SMS follow-ups on lead status progression.

Field	Console Location	Format & Description
Account SID	twilio.com/console > Dashboard	34-character string starting with AC.... Found under 'Account Info'.
Auth Token	twilio.com/console > Dashboard	Secret authentication token. Click 'Show' to copy.
Sender Phone Number	Phone Numbers > Active Numbers	SMS-enabled phone number in international E.164 format, e.g. +12025550199.
Inbound Webhook URL	Phone Numbers > Active > Messaging	Under 'A Message Comes In', select Webhook (HTTP POST) and enter: https://your-domain.com/api/v1/conversations/webhooks/twilio

4.4 Email Dispatch (Resend & Custom SMTP)

FlumenX supports two email delivery engines for outgoing lead communications, portal invites, and alert notifications:

Feature / Attribute	Option A: Resend (Recommended)	Option B: Custom SMTP
Setup Simplicity	Single API key (re_...). Minimal configuration.	Requires Host, Port, Username, Password, TLS settings.
Deliverability & DNS	Automated DKIM, SPF, and DMARC verification via Resend dashboard.	Requires manual MX, SPF, and DKIM configuration with email provider.
Supported Providers	Resend Cloud Engine.	Mailgun, SendGrid, Amazon SES, Google Workspace, Postmark, etc.
Configuration Fields	• API Key (re_...) • From Email (e.g. team@client.com) • Sender Name (e.g. Acme Sales)	• Host (e.g. smtp.mailgun.org) • Port (587 TLS or 465 SSL) • User & Password / App Password • From Email & Sender Name

5. Advertising & Lead Ingestion Setup

Connect advertising accounts under **Integrations Hub > Advertising & Ad Accounts** to synchronize daily campaign spend, impressions, clicks, CPC, ROAS, and capture incoming leads instantaneously.

5.1 Meta Marketing API (Facebook & Instagram Ads)

Required for syncing ad spend and automatically importing Facebook Instant Lead Form submissions directly into the CRM pipeline:

- **Ad Account ID:** Numeric ID prefixed with act_ (e.g. act_198273645019), found in Meta Ads Manager URL or Account Settings.
- **System User Access Token:** Generated in Meta Business Suite with the following required permissions:
 1. ads_read (Read ad campaigns, adsets, and daily spend performance metrics)
 2. leads_retrieval (Decrypt and download form field submissions)
 3. pages_show_list and pages_read_engagement (Identify form source pages)
- **Lead Gen Webhook:** In Meta App Dashboard > Webhooks > Select **Page**: Subscribe to leadgen. Whenever a user submits an instant form on Facebook or Instagram, Meta sends a webhook payload to FlumenX, which auto-creates the CRM lead and executes welcome workflows.

Instant Lead Webhook Payload Structure (Inbound JSON)

```
{
  "object": "page",
  "entry": [{
    "id": "10293847561",
    "changes": [{
      "field": "leadgen",
      "value": {
        "leadgen_id": "981726354819",
        "page_id": "10293847561",
        "form_id": "382910485720",
        "created_time": 1758019200
      }
    }]
  }]
}
```

5.2 Google Ads API (MCC & Customer Accounts)

Synchronizes search, display, and YouTube ad campaign performance metrics into the client reporting analytics engine:

- **Google Customer ID:** 10-digit ID formatted as xxx-xxx-xxxx, found in the top-right corner of Google Ads console.
- **Developer Token:** Approved developer token generated in your Google Ads Manager Account (MCC) under **Tools & Settings > API Center**.
- **Google Cloud OAuth 2.0 Credentials:** Client ID and Client Secret generated in Google Cloud Console with the Google Ads API enabled.
- **Refresh Token:** Offline OAuth 2.0 refresh token generated via token exchange script or OAuth playground.

6. Local Development, Webhooks & Tunneling

External platforms (Meta for Developers, Twilio, Resend) deliver incoming messages and lead payloads via HTTP POST webhooks to your server. When running in a local development environment (localhost:5000), public external servers cannot route traffic to your private machine.

To test end-to-end inbound messaging and instant lead capture locally, establish a secure public HTTPS tunnel:

Option A: Using ngrok (Fastest Setup)

```
# Run a public HTTPS tunnel forwarding to local Express server port 5000:
npx ngrok http 5000

# Output will display a Forwarding URL, for example:
# Forwarding: https://8f92-203-110-44-12.ngrok-free.app -> http://localhost:5000
```

Option B: Using Cloudflare Tunnel (Free, No Rate Limits)

```
# Install cloudflared and start an instant ad-hoc tunnel:
cloudflared tunnel --url http://localhost:5000
```

Configuring Tunnel URLs in External Consoles:

Replace https://your-domain.com with your temporary forwarding URL:

Integration Target	Local Webhook Endpoint to Register in External Console
WhatsApp Cloud Inbound	https://<your-tunnel-url>/api/v1/conversations/webhooks/whatsapp
Instagram & Messenger	https://<your-tunnel-url>/api/v1/conversations/webhooks/meta
Twilio SMS Inbound	https://<your-tunnel-url>/api/v1/conversations/webhooks/twilio
Meta Lead Gen Webhook	https://<your-tunnel-url>/api/v1/conversations/webhooks/meta-leads

Verifying Webhook Challenge Handshake (cURL Test)

```
# Test Meta webhook verification challenge locally via cURL:
curl "http://localhost:5000/api/v1/conversations/webhooks/whatsapp?hub.mode=subscribe&hub.verify_token=flumen_x_wa_verify_2026&hub.challenge;=115820124"

# Expected HTTP 200 response:
115820124
```

7. Security Protocols & Access Control (RBAC)

7.1 AES-256-GCM Cryptographic Vault

FlumenX ConectOS enforces military-grade authenticated symmetric encryption on all stored API keys, refresh tokens, and authentication secrets using AES-256 in Galois/Counter Mode (GCM).

- **Encryption Procedure:** For each secret, a cryptographically secure 16-byte random Initialization Vector (IV) is generated. The ciphertext and a 16-byte authentication tag are computed and serialized alongside the IV.
- **Integrity Guarantee:** In GCM mode, any tampering or corruption of the encrypted database document causes decryption to fail immediately with an authentication error, protecting against ciphertext modification attacks.

7.2 Role-Based Access Control (RBAC) Matrix

Platform permissions are enforced at both the Express API routing layer and MongoDB query layer across three built-in roles:

Role Name	Scope & Authority	Integration Privileges
Super Administrator (super_admin)	Global cross-client access to all agency tenants, system audit logs, user provisioning, and RBAC matrix.	Can view, edit, configure, and delete integrations across all client workspaces. Can inspect DLQ error dead-letters.
Client Administrator (client_admin)	Full workspace authority over assigned client tenant, team members, CRM leads, and customer portal.	Can connect, test, and manage channel credentials for their own workspace. Cannot view other client tenants.
Client Staff (client_staff)	Operational specialist / media buyer privileges for managing leads, replying in inbox, and executing tasks.	Can reply through configured channels; cannot view or modify raw API keys or connection secrets.

WEBHOOK SIGNATURE VERIFICATION

Every incoming payload from Meta (X-Hub-Signature-256) and Twilio (X-Twilio-Signature) is cryptographically validated using HMAC-SHA256 against the client's configured App Secret before processing. Forged or unsigned payloads are rejected with HTTP 403 Forbidden.

8. Troubleshooting & Diagnostics Guide

Consult this reference table for diagnosing common configuration, authentication, and webhook delivery errors:

Symptom / Error	Probable Root Cause	Corrective Action
HTTP 401: Unauthorized (Provider Test Failed)	Access token expired, revoked, or invalid permissions assigned.	Regenerate token in provider developer console (e.g. Meta System Users). Verify required scopes (e.g. whatsapp_business_messaging).
Webhook Verification Failed (Meta Dashboard Error)	Verify Token mismatch between Meta Console and FlumenX Webhooks tab.	Ensure the exact same token string is entered in both consoles. Verify server is reachable over public HTTPS URL.
HTTP 422: Unprocessable (Outbound Message)	Recipient phone number not formatted in E.164 standard or 24h WhatsApp service window closed.	Ensure phone numbers include country code (e.g. +12025550199). On WhatsApp, outside the 24h window, approved template messages are required.
CORS Origin Blocked (Browser Console Error)	CORS_ORIGIN in backend/.env does not match the frontend URL.	Update CORS_ORIGIN to match client URL (e.g. http://localhost:3000 or your live production domain). Restart backend server.
MongoDB Connection Refused	Local mongod service not running or Atlas IP whitelist blocking request.	In dev, ensure local MongoDB service is started (net start MongoDB). In Atlas, whitelist your server IP address.

Summary Checklist for Production Go-Live:

1. Generate cryptographically strong random keys for JWT_SECRET and ENCRYPTION_SECRET_KEY.
2. Change the Super Administrator initial password immediately after first login.
3. Switch MongoDB URI to an authenticated replica set with TLS and automated daily snapshots.
4. Register permanent System User tokens for Meta WhatsApp and Instagram channels.
5. Register public HTTPS webhook endpoints in Meta App Dashboard and Twilio Console.
6. Validate DNS records (SPF, DKIM, DMARC) for sending email domains.